

Formula Sheet

Gram Schmidt process

Given a basis $\{x_1, x_2, \dots, x_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}v_1 &= x_1 \\v_2 &= x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 \\v_3 &= x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 \\&\vdots \\v_p &= x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} v_2 - \dots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1}\end{aligned}$$

Then, $\{v_1, v_2, \dots, v_p\}$ is an **orthogonal basis** for W . An **orthonormal basis** is constructed from an orthogonal basis $\{v_1, v_2, \dots, v_p\}$ by normalizing all v_k , $1 \leq k \leq p$.

Classification of PDEs

- $Au_{xx} + 2Bu_{xy} + Cu_{yy} + F(\dots) = 0$
- Elliptic: $B^2 - AC < 0$
- Parabolic: $B^2 - AC = 0$
- Hyperbolic: $B^2 - AC > 0$

Standard PDEs

- Laplace: $u_{xx} + u_{yy} = 0$
- Poisson: $u_{xx} + u_{yy} = f(x, y)$
- Heat: $u_t = cu_{xx}$
- Wave: $u_{tt} = c^2 u_{xx}$

Discretization Formulas

- Laplace: $u_{i,j} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})$
- Poisson: $u_{i,j} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}) - \frac{h^2}{4} f_{i,j}$
- Heat: $u_{i,j+1} = \frac{u_{i+1,j} + u_{i-1,j}}{2}$
- Wave: $u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}$

Dijkstra's algorithm

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DIJKSTRA( $G, w, s$ )
1  INITIALIZE-SINGLE-SOURCE( $G, s$ )
2   $S = \emptyset$ 
3   $Q = G.V$ 
4  while  $Q \neq \emptyset$ 
5       $u = \text{EXTRACT-MIN}(Q)$ 
6       $S = S \cup \{u\}$ 
7      for each vertex  $v \in G.Adj[u]$ 
8          RELAX( $u, v, w$ )
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Conditional Probability

- If E and F are two events associated with the same sample space of a random experiment, the conditional probability of the event E given that F has occurred, i.e. $P(E|F)$ is given by

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \text{ provided } P(F) \neq 0$$

Bayes Theorem

Let $E_1, E_2, \dots, E_n$ be a set of events associated with a sample space S , where all the events $E_1, E_2, \dots, E_n$ have non-zero probability of occurrence and they form a partition of S . Let A be any event associated with S , then according to Bayes theorem,

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{k=1}^n P(E_k)P(A|E_k)}$$

Marginal Distributions

The **marginal distributions** describe the individual behavior of X or Y .

Discrete:

$$p_X(x) = \sum_y p(x, y), \quad p_Y(y) = \sum_x p(x, y)$$

Expectation and variance of two-dimensional random variables

$$E[X] = \sum_i x_i p_X(x_i), \quad E[Y] = \sum_j y_j p_Y(y_j)$$

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

t-statistics for two sample t test:

Used when: Population variances are unknown but assumed equal.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled standard deviation:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Degrees of freedom:

$$df = n_1 + n_2 - 2$$

t-table

TABLE VIII : CRITICAL VALUES OF STUDENT'S *t*-DISTRIBUTION

<i>d.f.</i>	LEVEL OF SIGNIFICANCE FOR ONE-TAILED TEST						
	.25	.10	.05	0.025	.01	.005	.0005
<i>v</i>	LEVEL OF SIGNIFICANCE FOR TWO-TAILED TEST						
	.50	.20	.10	.05	.02	.01	.001
1	1.000	3.078	6.314	12.706	31.821	63.657	636.619
2	.816	1.886	2.920	4.303	6.965	9.925	31.599
3	.765	1.638	2.353	3.182	4.541	5.841	12.924
4	.741	1.533	2.132	2.776	3.747	4.604	8.610
5	.727	1.476	2.015	2.571	3.365	4.032	6.869
6	.718	1.440	1.943	2.447	3.143	3.707	5.959
7	.711	1.415	1.895	2.365	2.998	3.499	5.408
8	.706	1.397	1.860	2.306	2.896	3.355	5.041
9	.703	1.383	1.833	2.262	2.821	3.250	4.781
10	.700	1.372	1.812	2.228	2.764	3.169	4.587
11	.697	1.363	1.796	2.201	2.718	3.106	4.437
12	.695	1.356	1.782	2.179	2.681	3.055	4.318
13	.694	1.350	1.771	2.160	2.650	3.012	4.221
14	.692	1.345	1.761	2.145	2.624	2.977	4.140
15	.691	1.341	1.753	2.131	2.602	2.947	4.073
16	.690	1.337	1.746	2.120	2.583	2.921	4.015
17	.689	1.333	1.740	2.110	2.567	2.898	3.965
18	.688	1.330	1.734	2.101	2.552	2.878	3.922
19	.688	1.328	1.729	2.093	2.539	2.861	3.883
20	.687	1.325	1.725	2.086	2.528	2.845	3.850
21	.686	1.323	1.721	2.080	2.518	2.831	3.819
22	.686	1.321	1.717	2.074	2.508	2.819	3.792
23	.685	1.319	1.714	2.069	2.500	2.807	3.768
24	.685	1.318	1.711	2.064	2.492	2.797	3.745
25	.684	1.316	1.708	2.060	2.485	2.787	3.725
26	.684	1.315	1.706	2.056	2.479	2.779	3.707
27	.684	1.314	1.703	2.052	2.473	2.771	3.690
28	.683	1.313	1.701	2.048	2.467	2.763	3.674
29	.683	1.311	1.699	2.045	2.462	2.756	3.659
30	.683	1.310	1.697	2.042	2.457	2.750	3.646
40	.681	1.303	1.684	2.021	2.423	2.704	3.551
60	.679	1.296	1.671	2.000	2.390	2.660	3.460
120	.677	1.289	1.658	1.980	2.358	2.617	3.373
∞	.674	1.282	1.645	1.960	2.326	2.576	3.291